F

Forage

Data Visualisation Empowering Business with Effective Insights

TATA

Mar 2026

Project completed as part of a Data Analytics Virtual Experience Program (Forage).

01

Overview

To successfully complete this analysis project, a structured data analytics workflow was followed. The process began with collecting and preparing the data, followed by cleaning and transforming the dataset to ensure accuracy. Finally, meaningful visualizations were created to generate insights that can help the CEO and CMO make strategic business decisions.

2026

Data Collection

2026

2026

This systematic approach ensures that the insights derived from the dataset are reliable, accurate, and aligned with the business objectives of the organization.

The first step of the project involved collecting the dataset and importing it into Power BI for analysis.

The dataset represents online retail transactions, where each row corresponds to a product purchase made by a customer. The dataset contains detailed information about products, customers, sales quantity, and pricing.

TATA

Key attributes available in the dataset include:

01

Invoice Number – Unique identifier for each transaction

02

Stock Code – Product identification code

03

Product Description – Name of the product purchased

04

Quantity – Number of units purchased in a transaction

Importing the dataset into Power BI allowed the data to be structured, explored, and prepared for further analysis.

2026 01 Invoice Date – Date when the transaction occurred

2026 02 Unit Price – Price of a single unit of the product

2026 03 Customer ID – Unique identifier for each customer

2026 04 Country – Country from where the order was placed

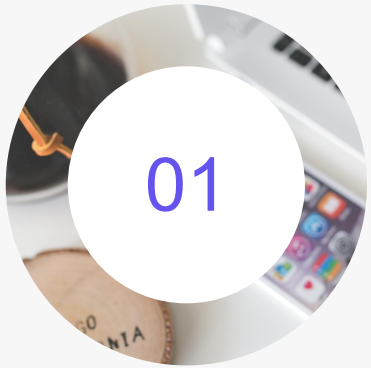
02

Data Cleaning and Preparation

Before performing any analysis, it was essential to ensure that the dataset was clean and free from incorrect or misleading records. Poor data quality can significantly affect the accuracy of insights and lead to incorrect business decisions.

2026

During the data review process, several issues were identified:



Some transactions contained negative quantities, which represent product returns.

Some records contained invalid unit prices.

Certain rows had missing customer identifiers.

A few records had incorrect or inconsistent values.

To address these issues, the following cleaning checks were applied:

These cleaning steps helped eliminate bad or inaccurate data, ensuring that the dataset used for analysis represented actual sales transactions.

01

Removed records where Quantity was less than 1, as negative quantities represent returned products rather than sales.

02

Removed records where Unit Price was less than 0, since a product price cannot be negative.

03

Removed rows where Customer ID was missing, as these records cannot be linked to a specific customer.

04

Verified data consistency to ensure all values were logically valid.

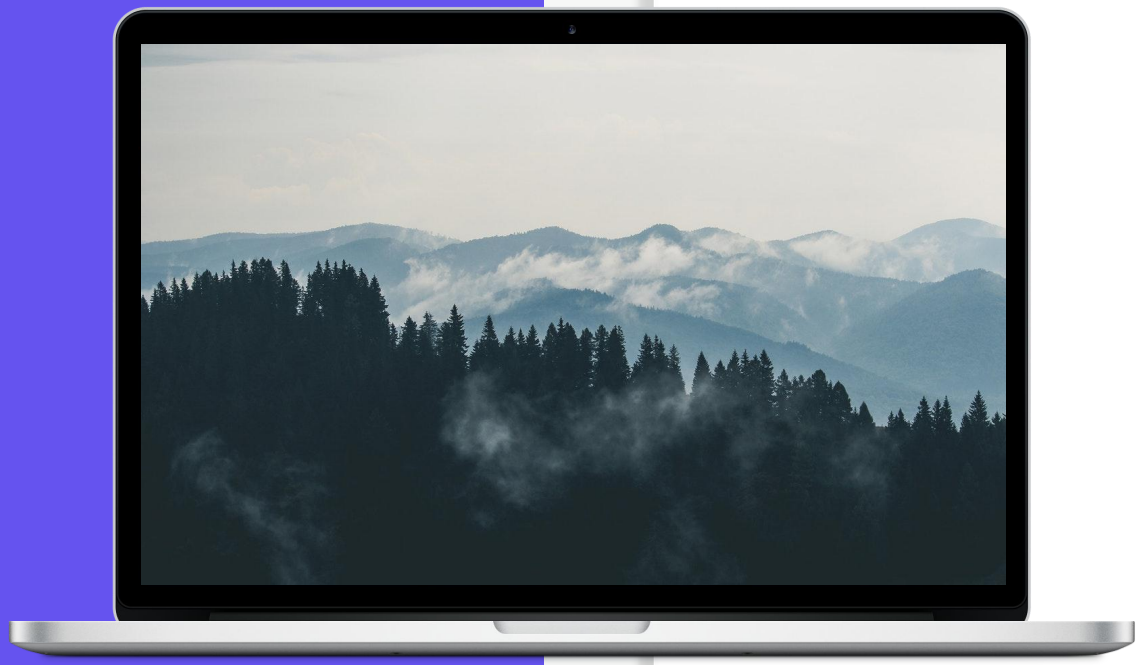
03

Data Transformation and Feature Engineering

Once the dataset was cleaned, the next step involved transforming the data to make it more useful for analysis.

2026

Data Transformation and Feature Engineering



A new calculated field was created to measure the total revenue generated from each transaction.

Revenue was calculated using the formula:

Revenue = Quantity × Unit Price

This calculation allowed the dataset to clearly represent the financial value of each sale, which is essential for analyzing business performance.

2026

Additional transformations were also performed to support the required analysis:

Extracted month and year information from the Invoice Date for time-based analysis.

Structured the dataset to enable country-level aggregation.

Prepared the data to identify top customers and top-performing countries.

These transformations allowed the dataset to be organized in a format suitable for business intelligence and visualization.

04

Data Visualization and Analytical Approach

After preparing the dataset, the next step was to create interactive visualizations using Power BI. The purpose of these visualizations was to answer the key business questions raised by the CEO and CMO.

2026

Each visual was designed to highlight specific aspects of the business performance.

The following visualizations were created:

30%

Line / Area Chart

- This chart was used to analyze monthly revenue trends for the year 2011.
- It helped identify seasonal patterns and fluctuations in sales throughout the year.

60%

Line & Clustered Column Chart

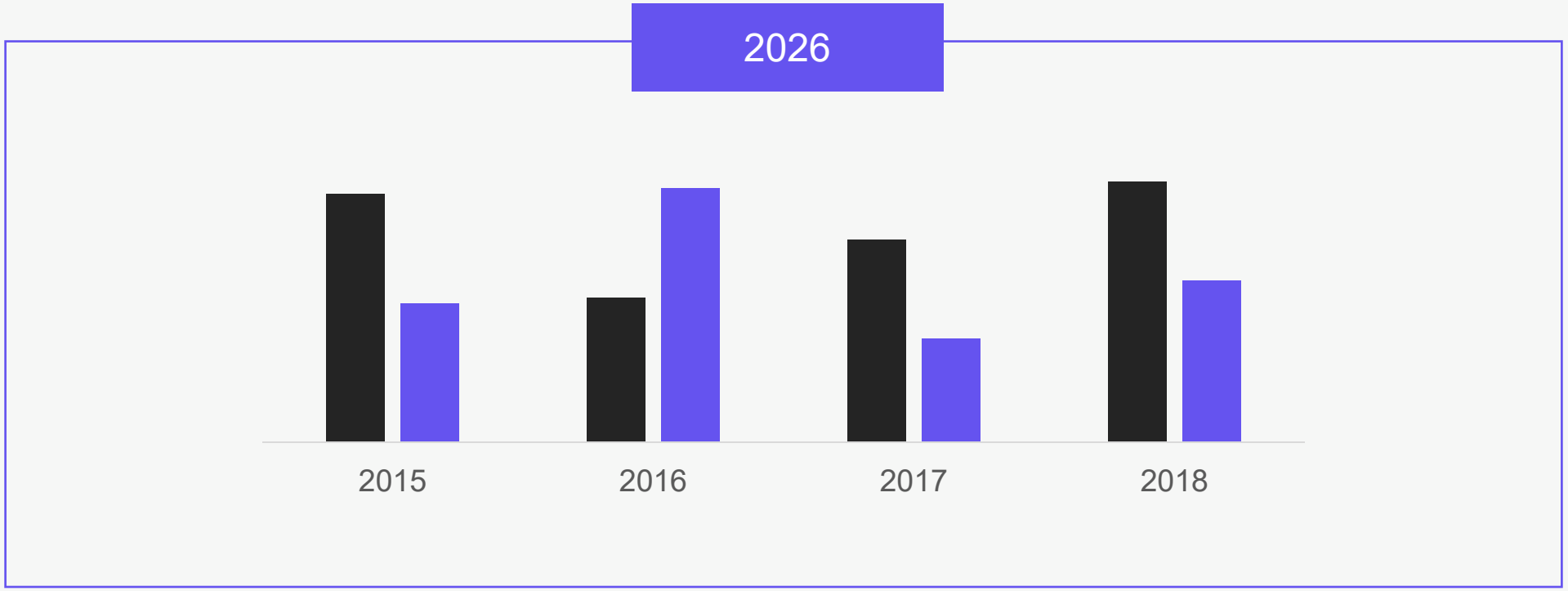
- This visualization was used to compare revenue and quantity sold across the top 10 countries.
- The United Kingdom was excluded to focus on international market opportunities.

30%

Treemap Visualization

- The treemap was used to identify the top 10 customers generating the highest revenue.
- This visual highlights the contribution of each high-value customer to total sales.

Column & Ribbon Chart



- This chart was used to analyze product demand across different countries.
- It helps identify which regions have the strongest demand for the company’s products.

These visualizations made the data easier to interpret and allowed executives to quickly understand patterns, trends, and performance drivers.

05

Business Insights and Strategic Decision Support

The final stage of the project involved interpreting the visualizations and extracting meaningful insights that could help the company make informed decisions.

2026

The analysis revealed several important findings:

2026 01 Sales demonstrate clear seasonal patterns, with higher revenue during the later months of the year.

2026 02 Several countries outside the United Kingdom generate significant revenue, indicating potential markets for expansion.

2026 03 A small group of customers contributes a large portion of the company's revenue, highlighting the importance of customer retention strategies.

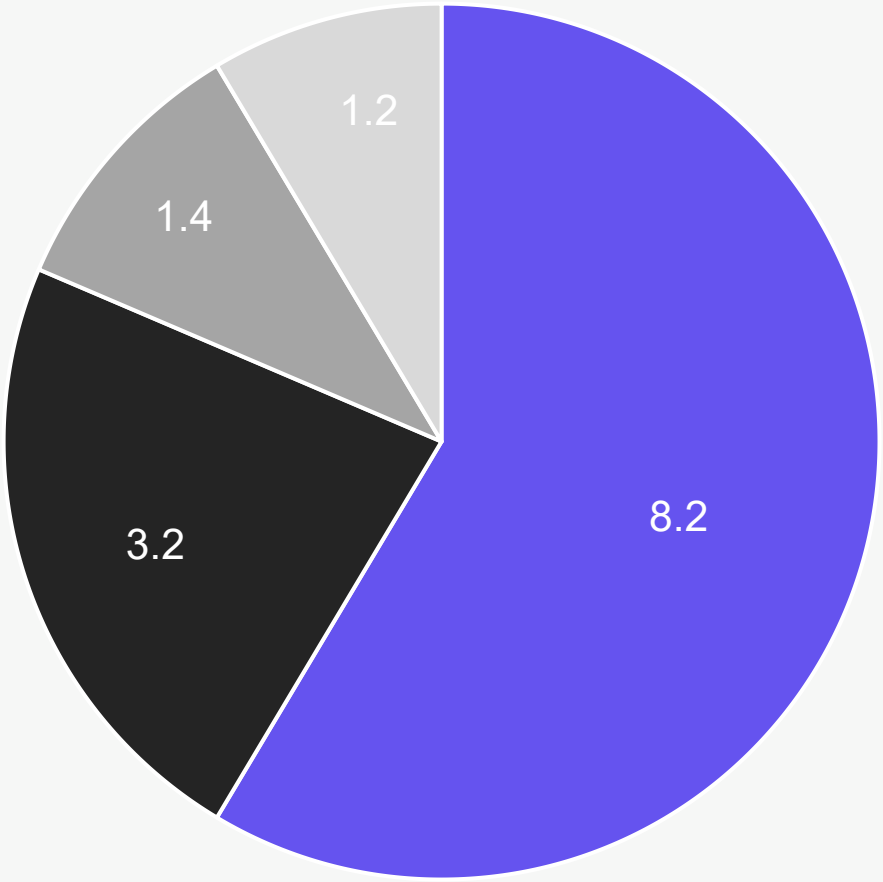
2026 04 Demand for products varies significantly across regions, suggesting opportunities for targeted marketing and inventory planning.

Business Insights and Strategic Decision Support

160$	**200$**	**320$**	**410$**

These insights provide valuable guidance for executives to improve market expansion strategies, customer engagement, and revenue growth.

These actions can help improve revenue growth and market reach.



01 Market expansion

02 Customer retention

03 Revenue growth

04 Seasonal sales planning

F

Forage

THANK YOU

TATA

2026